Reading Unseen Scripts: A Zero-Real-Image Evaluation Protocol and a Scored Forecasting Campaign for Brahmic Scene Text

Anonymous WACV Datasets Track submission

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Abstract

Scene-text recognition is effectively solved for Latin script and fails for most of the world's writing systems, yet no benchmark scores the case that decides the long tail: a script for which no real labeled image exists at all. We contribute a protocol for that case and the infrastructure to audit it — not a new dataset. It is leave-one-script-out over nine Brahmic scripts of a public benchmark: the held-out script appears only as rendered synthetic images, never a photograph, and is read in a round-trip-lossless shared grapheme pivot space, so pivot accuracy is target-script accuracy. An out-of-benchmark rung extends it to Khmer, a disjoint Unicode block. We release the pivot construction, the rendering pipeline, every split and result, and a harness that re-derives all 158 numbers here and fails on mismatch; a tenth script needs only a correspondence map and a font.

Applied to a compact open vision–language model, the protocol makes every held-out script readable — 9.16–30.09% word recognition from source-only baselines of 0.0–5.46% — and scales it against three reference systems on the identical crops under the identical metric: stock per-script Tesseract, trained on real target-script images, loses on eight of nine (19.41% vs. 14.35%); a prompted 7B VLM loses on eight of nine; the benchmark authors' supervised specialists reach 66.26%, of which we recover 0.29. Swapping the pivot for a stock BPE tokenizer under identical data, seed and recipe degrades all nine scripts (sign test $p = 0.0039$), isolating output-space structure as the mechanism.

Every forecast was committed before the run it predicts, and scoring them is what makes such an apparatus informative rather than decorative — it rules things out. The preregistered primary, that tokenizer fertility predicts transfer, is falsified ($\rho = -0.80$); a filed dose–response is log-linear with a script-invariant slope but missed its committed magnitude by 5.1×; Khmer, forecast before any Khmer recognizer existed, returned 0.8%, a -3.66σ miss that locates the method's boundary. We advance no predictive law — we advance the protocol that refuted these claims, the infrastructure to check it, and a scored record of what it ruled out.

1. Introduction

Modern scene-text recognizers exceed 90% word accuracy on Latin benchmarks, yet most of the world's writing systems remain unreadable to them. Recent audits are stark: across more than a hundred Unicode scripts, even frontier vision–language models (VLMs) read fewer than thirty, and on unfamiliar scripts they emit garbled or substituted output [15]. For the Brahmic family — nine major scripts used by over a billion people — real-image training data exists only because of recent, labor-intensive benchmark efforts [9, 20], and dozens of related scripts (Sinhala, Khmer, Myanmar, Tibetan, …) have little or none. Collecting and annotating real photographs script-by-script is not a scalable path to coverage — and, crucially, no existing benchmark measures how well a recognizer does when a script's real labeled images are absent entirely. That gap is an evaluation gap before it is a modeling one.

This paper is about closing it. We ask a question existing protocols cannot score — can a recognizer read a script for which no real labeled image exists at all? — and build the machinery to answer it honestly: a protocol isolating the zero-real-image condition, a forecasting discipline that commits every prediction before its run, and a verification chain that lets a reader re-derive every number. The protocol is leave-one-script-out (LOSO) over all nine scripts of a public real-image benchmark [9]: for each held-out script we train a source-only model (Rung A) and a source+synthetic-target model (Rung B), both evaluated on the held-out script's real test images. A further out-of-benchmark probe holds out Khmer, whose Unicode block and coeng-based stacking sit outside the nine; it returns 0.8% WRR against a 16.1% forecast filed before any Khmer recognizer existed, locating the protocol's boundary as sharply as its successes. The recognizer exercising this machinery is a compact open VLM (Florence-2, 0.23B [25]) fine-tuned in a shared grapheme pivot space: Brahmic scripts share an organizational scheme (consonant bases, vowel signs, virama-formed conjuncts), so each script's text decomposes into grapheme clusters mapping by structural correspondence into one output vocabulary. The pivot is the object the evaluation measures, not the headline claim.

Our contribution class is an evaluation protocol plus verification infrastructure, not a data release: we collect no new photographs, instead repurposing two public benchmarks [9, 22] under a new held-out condition and adding verified synthetic renderings. This is squarely what this track solicits — new evaluation protocols, rigorous auditing and stress-testing, and negative results — which it invites independently of new datasets.

Our contributions, in the order this paper weights them:

1. A zero-real-image evaluation protocol. A LOSO scene-text protocol that measures reading an entire writing system — disjoint Unicode block, disjoint glyph inventory — from zero real target images, in a shared pivot space with a round-trip-lossless mapping (so pivot-WRR equals target-script WRR), plus an out-of-benchmark generalization probe (Khmer). It surfaces a strong effect existing protocols never test: every held-out script moves from a $\leq 5.46\%$ source-only baseline to 9.16–30.09% WRR with zero real target images.
2. Prospective prediction as an evaluation practice — and a mostly negative record. A frozen, version-controlled preregistration with committed-before-run forecasts, and a discipline of scoring every call including its misses. We advance no predictive law — the record is the contribution, and it is largely one of failures. The preregistered primary (tokenizer fertility predicts transfer) was falsified (Spearman -0.80); a prospectively filed synthetic-exposure study gives a script-invariant, log-linear dose–response (+2.235 WRR/doubling, F-test $p = 0.45$) whose pre-committed magnitude missed by 5.1×; pivot coverage orders scripts in-sample but not out-of-sample ($r = +0.03$). Every point forecast is auditable, hits and misses alike (Kannada $\sim 15 \rightarrow 15.42$; Gurmukhi 16.2 filed 3.7 h pre-training $\rightarrow$ 22.16; Khmer 16.1 $\rightarrow$ 0.8, a -3.66σ miss).
3. Controlled attribution of the effect — the pivot is the mechanism. A preregistered stock-BPE ablation under identical data, seed, and recipe degrades WRR on all nine scripts (1.08×–3.91×; sign test $p = 0.0039$), isolating output-space structure, not synthetic fine-tuning per se, as the driver; a from-scratch CRNN replicates the direction across an architecture with no large-scale pretraining but not the magnitude, and we report the ambiguity rather than resolve it in our favor.
4. System analysis on the benchmark, and released resources. On the identical test crops we situate a prompted 7B frontier VLM (reads two of nine; loses to the compact model on eight of nine), off-the-shelf per-script OCR (14.35%, which our zero-real-image model beats on eight of nine), and the benchmark authors' own supervised specialists (66.26%, of which we recover 0.29) — an interpretable floor-to-ceiling context — alongside the released protocol, harness and preregistration chain (Sec. 3.3).

2. Related Work

Multilingual STR and Indic benchmarks. STR is mature for Latin script [4] but collapses elsewhere. IndicSTR12 [20] and the Bharat Scene Text Dataset [9] established real-image benchmarks for Indian languages; on the latter, PARSeq reaches ∼47% average WRR from synthetic data alone and ∼73% fine-tuned on real labeled images — numbers that presuppose labeled target data. GlotOCR [15] quantifies the wider problem: frontier VLMs read fewer than thirty of 100+ scripts. Synthetic-data engines remain Latin-centric [26], and synthetic-rendering fine-tuning has been shown for a single low-resource script with real-data validation [7]. We target the regime where no real labeled target image exists at all.

Tokenization granularity for complex scripts. MGP-STR [8] fuses character, BPE and WordPiece streams in one Latin recognizer; grapheme tokenization beats byte-BPE for Bengali handwriting [12]; and fertility's value as a tokenizer metric is contested in the LLM literature [21]. A companion manuscript by the same authors [1], anonymized in the supplement, makes the supervised grapheme-vs-BPE choice predictable across nine scripts (polarity 8/9, leave-one-script-out 90.9%); the present paper shares no experiment with it and tests — and refutes — the conjecture that the same property predicts zero-real-image transfer.

Unseen units vs. unseen scripts. A substantial line reads unseen characters within a known script by de-

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composition into shared sub-units: stroke- and radical-based zero-shot Chinese recognition [5, 27, 28] and open-set recognition of novel characters [17–19]. Zero-shot detection of unseen scripts localizes but does not read [16]. Byte-level pivots let ASR cover unseen-script languages [6], and task-arithmetic analogies achieve zero-real-data synthetic-to-real HTR for five Latin-script languages, leaving unseen writing systems explicitly open [11]. To our knowledge no prior system reads real scene text of an entirely unseen script — different Unicode block, different glyphs — from zero real target images.

What predicts cross-lingual transfer. In NLP, transfer tracks lexical and subword overlap rather than language relatedness [10, 23]. For STR specifically, Ref. [2] concludes that source dataset size and visual similarity, not linguistic typology, drive cross-lingual transfer. Our findings sharpen rather than contradict this, in a regime that work did not study (zero real target images): (i) a typology-adjacent structural property — fertility — indeed fails as a transfer predictor ($\rho = -0.80$), as a data-centric view expects; (ii) synthetic target exposure dominates — a target-side data-size effect; (iii) within a fixed budget, coverage of the shared grapheme space orders transfer ($\rho = +0.61$ among equal-budget scripts) — an OCR instantiation of the subword-overlap principle in a designed output space; and (iv) a frozen-encoder visual-similarity control predicts nothing in our data ($\rho = -0.12$). Because LOSO balances source data by construction, the confound identified by [2] is controlled by design.

Prediction and preregistration in empirical ML. Scaling laws make supervised STR predictable in model and data size [24]; preregistration and negative-result reporting are the reproducibility literature's recommended remedies for HARKing and cherry-picking [13, 14], yet full-study adoption is rare. We run the entire campaign under a frozen preregistration with an append-only amendment log and commit quantitative forecasts before the runs they predict; we offer the protocol itself as a contribution.

3. Method

3.1. A shared grapheme pivot space for Brahmic scripts

Brahmic scripts encode the same phonological skeleton — consonant bases, dependent vowel signs, and conjunct formation through a virama — and Unicode preserves this: corresponding letters sit at the same offset within each script's block. We exploit both layers.

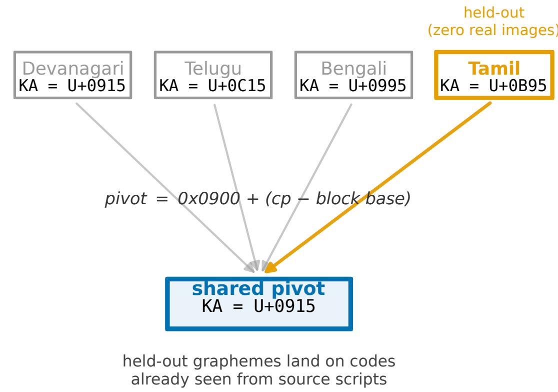

Figure 1. The shared grapheme pivot space. Each Brahmic script maps bijectively into a common output space by ISCII-derived Unicode-offset correspondence (the same letter sits at the same offset within each block); grapheme clusters in that space form the recognizer's output vocabulary. A held-out script's graphemes therefore land on pivot codes already seen from the source scripts. The held-out script enters training only as verified synthetic renderings (Rung B) or not at all (Rung A); evaluation is always on its real scene images.

Structural pivot. A mapping π sends any Brahmic codepoint to a pivot codepoint by its within-block offset ($\pi(c) = 0\text{x}0900 + (c - \text{base}(c))$); shared punctuation that is not offset-safe is excluded by construction. Because π is a per-script bijective shift, it is exactly invertible: a hypothesis decoded in pivot space is read back into the target script deterministically. Text from nine scripts thereby writes into one label space in which structurally corresponding syllables become identical strings.

Grapheme-cluster vocabulary. Within pivot space we segment text into grapheme clusters (consonant–virama conjuncts kept whole, dependent signs attached) and inject the resulting cluster inventory (1624 types for the source pool) as dedicated tokens into the recognizer's tokenizer. The alternative — letting the stock byte-BPE tokenizer segment pivot-space text — is the ablation of Sec. 4: it sees the same pivot strings but fragments clusters into sub-syllabic pieces. Fertility (mean clusters per word) governs when grapheme tokenization beats BPE in supervised training: in a 12-run balanced-budget study across the same nine scripts, fertility predicts the winning branch with 8/9 polarity and 90.9% leave-one-script-out direction accuracy ($R^2 = 0.678$). That result motivates both the pivot design and the (ultimately falsified) primary hypothesis of Sec. 5.

3.2. Zero-real-image protocol

For each held-out script S of the nine in the benchmark [9]:

Sources. All benchmark languages whose script is not S (languages sharing S, e.g. Assamese for Bengali script, are excluded — S is genuinely unseen). Each source language contributes exactly 700 train / 90 val real images (seed 42), giving balanced source exposure by construction; source-side data size is therefore controlled, unlike prior cross-lingual STR studies [2].

Synthetic target. 3240 images per target language: benchmark-vocabulary words rendered in fontconfig-verified fonts (multiple faces per script, libraqm shaping), each render checked non-blank; the rendered word enters training as its pivot transcription. No real photograph of S is ever seen. Scripts written by two source languages receive two languages' worth of renderings (2× budget) — a covariate disclosed at preregistration time that becomes central in Sec. 5.

Rungs. Rung A trains on sources only; Rung B adds the synthetic target set; Rung B_{bpe} repeats Rung B with the grapheme-cluster injection removed (stock tokenizer), identical data, seed, and recipe. All rungs fine-tune Florence-2 (0.23B) [25] with LoRA ($r = 16$, $\alpha = 32$, dropout 0.05), lr 10^{-4}, batch size 4, 15 epochs, validation-selected checkpoint, beam-search decoding at test.

Evaluation. The held-out script's real test split, WRR = exact match after NFC normalization, zero-width-character removal, and whitespace collapse — the benchmark's own protocol [9, 20] — plus character accuracy (1 − CER).

3.3. Preregistration and prospective predictions

The campaign ran under a frozen preregistration with an append-only amendment log; every amendment was filed before the results it concerns existed. Descriptors entering the transfer horse-race were computed result-blind: fertility and pivot coverage before any Rung-B result; a frozen-encoder visual-similarity control before the third. Quantitative forecasts for later runs were committed and pushed to a version-controlled archive before those runs trained, and are scored against a rule fixed at filing time — hits and misses alike (Sec. 5). We state the limit of this plainly, because it is the weakest link in the design. The archive is a version-controlled repository that was private during the campaign: no third party observed the commits as they were made, and git's own timestamps are settable by the author. The supplement therefore ships the filed documents themselves under SHA-256, which establishes that they are unaltered — not when they were written. At review time, filing order rests on our attestation; publishing the repository at camera-ready adds provenance but not proof, and we do not offer it as proof. A campaign starting today should anchor each filing, at the moment it is filed, in a timestamp no author controls — an anonymized preregistration or a blockchain-anchored hash. Ours predates that decision, and we ask the reader to discount the prospectivity claim accordingly rather than to assume it.

Released resources. We release the apparatus, not a description of it: the pivot construction and its 1624-type grapheme vocabulary; the verified rendering pipeline; every LOSO split actually used (nine scripts × rungs A/B/B_{bpe}, plus the Khmer rungs); every per-run result file the paper cites, anchors included; the preregistration chain with each forecast and its scoring receipt; and verify_wacv_numbers.py, which re-derives every derivable number in this paper from those result files and exits non-zero on any mismatch (158 macros, zero mismatches at submission). Extending the protocol to a tenth script needs only a correspondence map and a font — the Khmer rung is the worked example, and the harness is unchanged.

4. Experiments

The comparisons that match our question are the source-only baseline (Rung A), the raw pre-trained VLM (WRR 0.0), the stock-BPE ablation, the off-the-shelf, frontier-VLM and supervised-specialist references of Sec. 4.2, and our own forecasts.

4.1. Main result: every unseen script becomes readable

Table 1 is the campaign's core. Every one of the nine scripts moves from a near-floor source-only baseline (0.0–5.46% WRR; the raw pre-trained model reads none of them) to 9.16–30.09% WRR with character accuracy 34.57–57.77% — without a single real target image. Transfer does not collapse even on the script with least to borrow (Malayalam, lowest pivot coverage at 79.2%: still 0.18 → 9.69). Rung-A baselines track incidental pivot-space coverage (highest for Devanagari, 5.46%), confirming the pivot shares structure before any target exposure.

4.2. Floor and ceiling

Absolute numbers need a scale, and the supervised ceiling alone is the wrong one. Against the ceiling we recover 0.29 of the macro-average (19.41% vs. 66.26%): real labels remain worth roughly 3.4× what our synthetic pivot buys, and we do not present that as close. Against the floor the picture inverts. Stock Tesseract

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Script	N	A	B	B 95% CI	CharAcc	B_{bpe}
Tamil	513	0.0	9.16	[6.6, 11.7]	39.0	2.34
Telugu	545	1.28	19.82	[16.5, 23.1]	54.43	10.28
Kannada	720	2.78	15.42	[12.8, 18.1]	46.37	13.33
Malayalam	547	0.18	9.69	[7.3, 12.2]	34.57	4.2
Oriya	1044	0.77	25.38	[22.7, 28.1]	56.37	14.94
Gujarati	1015	3.05	15.86	[13.6, 18.2]	38.49	13.69
Bengali	2873	1.39	27.08	[25.5, 28.7]	57.77	19.07
Devanagari	6042	5.46	30.09	[28.9, 31.3]	57.16	27.99
Gurmukhi	2879	0.59	22.16	[20.7, 23.7]	44.1	17.65
macro-average			19.41			

Table 1. Zero-real-image transfer across nine held-out Brahmic scripts (real test images). Rung A: source scripts only. Rung B: + synthetic target renderings, zero real target images (95% CI = percentile bootstrap over test samples, 5000 resamples). B_{bpe}: Rung B with the grapheme-cluster vocabulary removed (stock tokenizer), identical data and seed. On 6/9 scripts the Rung-B CI lies strictly above the B_{bpe} 95% CI. Reference systems on these same crops are in Table 2.

Script	Tess.	Qwen	Rung B	PARSeq	o/c
Tamil	15.4	6.24	9.16	76.61	0.12
Telugu	13.76	0.18	19.82	55.78	0.36
Kannada	12.08	0.28	15.42	59.44	0.26
Malayalam	5.85	0.0	9.69	64.53	0.15
Oriya	15.52	0.38	25.38	67.62	0.38
Gujarati	12.41	0.3	15.86	57.04	0.28
Bengali	15.32	24.82	27.08	72.82	0.37
Devanagari	19.78	35.05	30.09	68.9	0.44
Gurmukhi	19.03	0.52	22.16	73.6	0.30
macro-avg	14.35	7.53	19.41	66.26	0.29

Table 2. Every reference system on the identical crops, through the identical pivot mapping and exact-match metric. Tess. — stock per-script Tesseract 5, trained on real labeled images of each target script (off-the-shelf floor). Qwen — Qwen2.5-VL-7B prompted zero-shot, scored under the disclosed most-favorable extraction. PARSeq — the benchmark authors' own supervised specialists, also trained on real labeled target images; BSTD train and test share no source image on any script, and the measured mean is in line with the ∼73% they report. o/c = ours / ceiling. Both trained-on-real systems presuppose exactly the data whose absence defines our setting.

ships a per-script model for every script here, trained on real target images, and is what a practitioner deploys today when a script is not otherwise served: it reaches 14.35%, and our zero-real-image model is ahead on eight of nine, and ahead by +5.06 points on macro-average, losing only Tamil. Tesseract is a document engine outside its design domain, so this is no claim to be the better recognizer in general — it is evidence that the zero-real-image regime has already passed what is obtainable off the shelf. Both hold at once: far from what real labels buy, past what is deployable without them.

4.3. The pivot is the mechanism: BPE ablation

Removing the grapheme-cluster vocabulary — same pivot strings, same images, same seed — degrades all nine scripts (Table 1), from 1.08× (Devanagari, 30.09 → 27.99 WRR) to 3.91× (Tamil, 9.16 → 2.34): the margin varies, the direction never does (exact two-sided sign test, $p = 0.0039$). The bootstrap makes the per-script margin concrete: on 6 of nine the Rung-B 95% CI clears the B_{bpe} interval outright (Table 1); the three exceptions — Devanagari, Kannada, Gujarati — are exactly the near-parity scripts (1.08–1.16×), consistent with the direction holding but the magnitude shrinking as fertility falls. The character-accuracy pattern is diagnostic: on Telugu, Kannada and Devanagari the BPE model equals or exceeds the pivot model's character accuracy (Telugu 55.1% vs. 54.43%), yet word accuracy drops in every case. Synthetic fine-tuning alone teaches the model to see the characters; the cluster-level output space is what assembles them into correct whole words. The margin itself is not noise: it grows with the held-out script's fertility, largest exactly on the high-fertility Dravidian scripts where transfer is hardest (Sec. 5.1).

4.4. Dose–response: synthetic budget

We hold the recipe fixed and vary only the number of synthetic target images, for the three scripts whose held-out budget we can grow — Malayalam, Kannada, Telugu — across $\{810, 1620, 6480, 12960\}$ renderings; the external Rung-B point (3240) anchors each curve. Word recognition is cleanly log-linear in synthetic exposure (R^2 from 0.905 to 0.999 per script), and the slope is script-invariant: a single shared +2.235 WRR per doubling fits all three, with no evidence of differing slopes (F-test $p = 0.45$; Fig. 2). This is the preregistered form, and it holds.

The preregistered magnitude does not. The frozen instrument projected +11.48 WRR per doubling — transported from the in-script 2×-synthetic experiment that fit the two-factor model (Sec. 5) — which the swept slope excludes by a factor of 5.1: the faithful doubling step (3240 → 6480) gained +2.35 WRR on average against a predicted +11.5, a miss outside the ±2 RMSE band and reported as filed (open squares, Fig. 2). Two further preregistered edges resolve the same way. The low-budget prediction of near-collapse toward the Rung-A floor is falsified — Kannada and

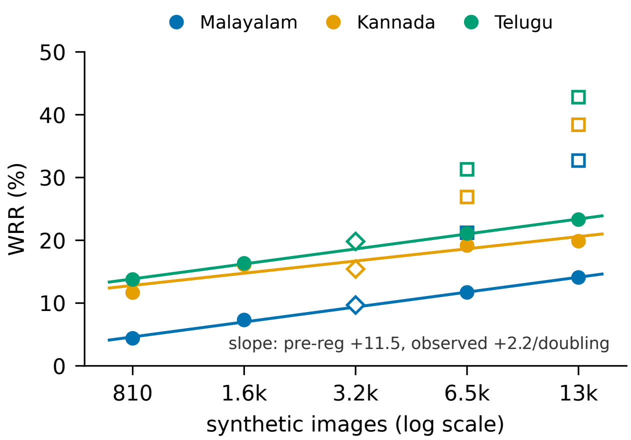

Figure 2. Synthetic-budget dose–response for three held-out scripts. Filled circles: observed WRR on real test images; solid lines: per-script log-linear fits (shared slope +2.235/doubling); open diamonds: the external Rung-B anchor (3240 images); open squares: the preregistered point predictions — their gap above the observed points is the 5.1× magnitude miss, reported as filed.

Telugu already reach double digits at 810 images, so transfer is concave, not linear, in log-budget — while the top step (6480 → 12960), which adds renderings at near-fixed lexicon, gains little on Kannada (onset of lexicon saturation) but keeps pace on Malayalam and Telugu. Synthetic scaling is thus real and orderly, but far shallower than an in-script extrapolation predicts; we report the confirmed form and the missed coefficient together. We do not promote pivot-space coverage to a cross-script driver: it orders the three swept scripts in-sample, but across the six scripts never used in the fit it is uncorrelated with transfer ($r = +0.03$), so the coverage axis (three distinct values) is underdetermined and we disclose it rather than build on it.

4.5. Architecture generality: a preregistered CRNN replication

Every result above uses one backbone, inviting the objection that the effect is specific to a pretrained vision–language model. We preregistered a replication (Amendment 6, filed and committed before any CRNN trained) on a from-scratch CRNN — ResNet+BiLSTM, CTC, no large-scale pretraining — over the same LOSO splits, pivot space, and metric, for the three equal-synthetic scripts spanning the observed Rung-B range (Tamil, Telugu, Oriya), with the vocabulary built from TRAIN records only and a 60-epoch budget fixed in advance. We filed four predictions and score all four, hits and misses alike.

The two structural predictions hold. Source-only Rung A stays at the floor on all three (WRR ≤ 0.67; predicted < 2.0), and the held-out ranking is preserved ($0.97 < 1.1 < 4.5$, matching the Florence-2 order, Spearman $+1$); the direction of the effect replicates, with Rung B above Rung A on every script. The two effect-size predictions miss. The preregistered primary — a $\geq +2.0$ WRR lift on at least two of three — holds only on Oriya (+3.83); Tamil (+0.97) and Telugu (+1.1) fall short. The CRNN reaches only a 0.11× fraction of the Florence-2 transfer level on average (predicted band $[0.15, 0.60]$), landing inside the band on Oriya alone (0.18). Character accuracy is depressed in step (e.g. Tamil 25.09% vs. Florence-2's 39.0%): the small model sees fewer characters, not merely fewer whole words.

As stated in the preregistration's declared caveat, this outcome is ambiguous and we do not resolve it in our favor. An attenuated-but-directional replication on a small from-scratch model is consistent both with the pivot mechanism being partly dependent on the backbone's pretraining and with a plain capacity floor — and the depressed character accuracy supports the latter. We report the direction as replicated across architectures, leave the magnitude open, and keep the backbone-specificity limitation.

4.6. Out-of-benchmark: Khmer

Every script above shares the ISCII-aligned Brahmic design the pivot was built around. To find where the method stops, we ran it somewhere it should struggle: Khmer, a disjoint Unicode block with coeng-based stacking, outside the benchmark and outside India. Reaching it required a hand-curated extension of the pivot (94.41% of test codepoints map; the remainder are approximated or dropped, and counted as misses). Under Amendment 4b we hold Khmer out entirely and evaluate on 1252 KhmerST [22] word crops: Rung A trains on the nine benchmark scripts with no Khmer of any kind, Rung B adds 3240 synthetic Khmer words and still no real Khmer image. Before any Khmer recognizer existed we filed, committed and pushed a point prediction from the frozen two-factor model — 16.1 WRR, $\pm 1\cdot$RMSE $[11.9, 20.3]$— together with a written caveat that we expected it to be fragile.

It was. Rung B reaches 0.8% WRR against a predicted 16.1: a residual of -15.3, or $-3.66\cdot$RMSE, below even the $\pm 2\cdot$RMSE band $[7.8, 24.5]$. The miss is not an artifact of the coverage recheck disclosed at build time — re-evaluating the same instrument at the realized token coverage (88.01% rather than the filed 89.02%) moves the point only to 15.52. Only the structural prediction survives: Rung A sits at 0.0 (predicted < 5). The instrument had already lost its slope magnitude in block (Sec. 4.4); this is the first script we asked it

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to predict outside the blocks it was fit on, and it failed there too. We do not carry it forward as a contribution.

Two things nonetheless hold. The direction replicates: synthetic target-script exposure through the pivot beats the no-target-data control here as everywhere (+0.8 WRR, +5.85 character accuracy), so the mechanism is not inert out of block. But the magnitude collapses — 0.8 is 4.1% of the nine-script Rung-B mean and 8.7% of the weakest benchmark script. For calibration, stock Tesseract's Khmer-specific model, trained on real Khmer and scored on the identical crops through the identical pivot and metric, reaches only 2.16% here against 14.35% across the nine benchmark scripts: these crops are far harder than the benchmark for a supervised system too, and our zero-real-image model recovers 0.37 of that floor. We report this as context, not as a rescue — both numbers are near the floor, and 0.8% is a failure to transfer usefully.

We draw the boundary claim narrowly. Khmer received the same 1× synthetic budget and comparable nominal token coverage (88.01%) that yield double-digit WRR in block, and still returned 0.8. Volume and coverage were held up; typological proximity was not; the transfer collapsed. That is the cleanest evidence we have that relatedness, not synthetic volume, carries the effect — but it is a single script, and we state it as suggestive rather than decisive.

4.7. Frontier-VLM reference

We prompt Qwen2.5-VL-7B [3] zero-shot on the same nine real test sets (fixed prompt, greedy decoding), preregistered under Amendment 4c. Under the declared primary metric — exact word match on the raw generation, our shared normalization — it scores 0.0% on all nine: the chat model wraps every answer in prose ("The text in the image is …"). We therefore report a disclosed, most-favorable re-score (Table 2): we extract the first quoted span from each generation and apply the identical metric. Even so, the VLM reads only the two highest-resource scripts with any competence — Bengali 24.82% and Devanagari 35.05% — and is at or near zero on six of the remaining seven (at most 6.24%, on Tamil), frequently transcribing a wholly different script (Gurmukhi returned in Devanagari, Malayalam in Bengali). Our compact pivot model wins on eight of nine at a fraction of the parameters, the VLM leading only on Devanagari. The comparison is not like-for-like — the VLM's training data is unknown — but scale alone does not read these scripts; frontier models cover fewer than thirty of 100+ writing systems [15].

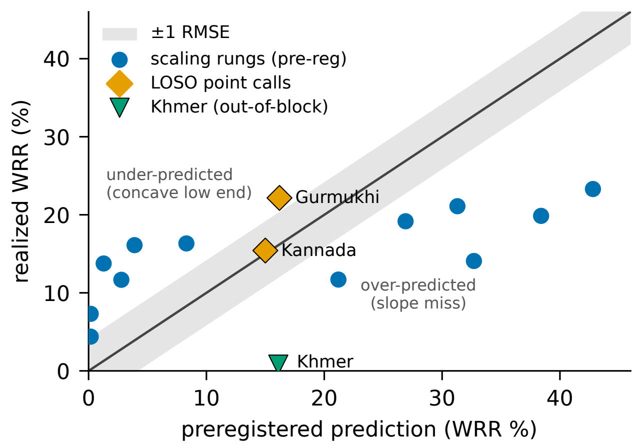

Figure 3. Prospective scorecard: preregistered prediction vs. realized WRR for every quantitative forecast filed before its run, scored against the rule fixed at filing time (shaded ±1 RMSE). The in-block LOSO point calls land on the diagonal; the scaling rungs fall off it both ways — under-predicted at low budget, over-predicted at high. Khmer, the out-of-block call, misses low by −3.66·RMSE. Hits and misses are both shown.

5. What Predicts Transfer?

We report the confirmatory analysis exactly as preregistered, our own primary hypothesis included.

5.1. The preregistered primary fails

Fertility — the property that governs the supervised grapheme-vs-BPE trade-off (Sec. 3.1) — was preregistered as the primary transfer predictor (H3). It is refuted: Spearman $\rho = -0.80$ across the nine Rung-B results — directionally opposite to the hypothesis. The flagship case was called in advance by the declared competitor: our prospective prediction file ranked Malayalam best under fertility (P1) and worst under coverage (P2); it realized bottom-tier (9.69 WRR). We report this as a genuine falsification, consistent with fertility skepticism in the LLM-tokenizer literature [21] and with the data-centric view of STR transfer [2].

Post-hoc, fertility instead tracks who needs the pivot — BPE comes closest to parity on the lowest-fertility script (Devanagari, 1.08×) and collapses on the highest (Tamil, 3.91×) — but that is an unregistered observation at $n = 9$, offered as a candidate for confirmatory testing and used for nothing here.

5.2. The two-factor account

Two result-blind quantities describe the campaign and generated its forecasts; we state them as description, not inference, because at $n = 9$ scripts nothing here carries appreciable inferential force. Synthetic budget dominates: the two scripts with the disclosed $2\times$ budget (Bengali, Devanagari) are the top two results (27.08, 30.09 WRR), Bengali from the second-lowest coverage. Coverage orders the seven equal-budget scripts ($\rho = +0.61$). The joint fit $\text{WRR} = -35.79 + 0.583 \cdot \text{cov} + 11.48 \cdot \mathbb{1}[2\times]$ (RMSE 4.18 WRR) is the instrument every later forecast was drawn from, so we print it for audit — and then discount it ourselves: a script-level bootstrap puts its coverage slope at $[-0.48, 3.49]$, spanning zero, and that term fails out-of-sample outright ($r = +0.03$, Sec. 4.4). Coverage is an in-sample ordering and nothing more. One control stands on its own: rendered glyphs embedded by the recognizer's own frozen vision encoder predict transfer not at all ($\rho = -0.12$), separating whatever the ordering reflects from the visual-similarity account of [2].

5.3. Prospective scorecard

Four forecasts were filed before their runs; all are scored (Fig. 3). Kannada: exploratory point ∼15 WRR, filed with two scripts observed; realized 15.42. Devanagari: directional call (high, ∼25+) committed before the result existed but pushed after — only weakly prospective, and we lean on it for nothing. Gurmukhi (final script): point 16.2 WRR, bands $\pm 1\sigma$ [12.2, 20.2] / $\pm 2\sigma$ [8.2, 24.1], committed and pushed 3.7 hours before training began; realized 22.16 — outside the $\pm 1\sigma$ band (under-called by 1.5σ), inside $\pm 2\sigma$. Khmer (Sec. 4.6), the one call filed for a script outside the pivot's Unicode blocks: point 16.1 WRR, realized 0.8 — a -3.66σ miss, below even the $\pm 2\sigma$ band. The declared rank bet between the two preregistered predictors resolved for coverage over fertility, neither ranking alone significant at $n = 7$.

The summary is bounded. In-block, transfer is forecastable to roughly ± 4.18 WRR (1σ) from two pre-training quantities — but even there the coverage term fails out-of-sample ($r = +0.03$) and the transported slope magnitude missed by $5.1\times$, and the single out-of-block call missed by -3.66σ. We report a scored, partly-failed forecast, not a contribution, and decline to extend it past the blocks it was fit on: one script shows it failed there, not that it must fail everywhere outside.

6. Limitations

Absolute accuracy. Our macro-average (19.41%) clears the off-the-shelf per-script OCR floor (14.35%), but the weakest script does not — Tamil (9.16%) falls below it — and all nine remain far below the supervised ceiling (∼73% with real labeled data [9]); the synthetic → real domain gap is the dominant residual, quantified by training loss reaching ∼0.06 on synthetic data while real-image WRR stays modest. One family, and a located boundary. Nine scripts, one benchmark, one family whose shared structure the pivot exploits by design. Khmer (Sec. 4.6) does not merely leave non-Brahmic generalization open — it shows the method failing there (0.8% WRR), and we make no claim beyond the family. One architecture. All transfer results use Florence-2; a from-scratch CRNN replicates the direction but at ∼0.11 of the magnitude (Sec. 4.5), so backbone-independence of the effect size is untested. Model imperfections. Gujarati underperforms its coverage, one Rung-A baseline breaches the predicted < 5 line (Devanagari, 5.46), and the final-script forecast under-called by 1.5σ; all reported, none explained away. Compute. Single 24 GB GPU; budgets (700/source, 3240 synthetic) were fixed by preregistration, not tuned.

7. Conclusion

We built an evaluation protocol for a condition existing benchmarks do not score — a script with no real labeled image at all — and ran a forecasting campaign inside it. Under that protocol, real scene text of an entirely unseen Brahmic script is readable at 9.16–30.09% word accuracy across all nine scripts of a public benchmark, ahead of stock per-script Tesseract on eight of nine despite never seeing a real image of the target, by a small open VLM fine-tuned in a shared grapheme pivot space with only rendered synthetic exposure; a stock-BPE ablation collapses that transfer, isolating output-space structure as the mechanism. The campaign ran under a frozen preregistration, and its forecasting record is mostly a record of misses: the primary hypothesis falsified, the scaling magnitude missed by $5.1\times$, coverage failing out-of-sample, and the one out-of-block forecast (Khmer) missed by -3.66σ. We advance no predictive law; every call ships in the supplement with its scoring receipt, under the provenance limits stated in Sec. 3.3, and the failures locate the method's boundary rather than leaving it for someone else to find. Beyond the specific results we offer the recipe — structural pivot, verified rendering, forecast-before-you-run — as a template for the long tail.

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